

Wei Xu

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EDUCATION

California Institute of Technology

Ph.D. Student in Applied Physics; M.S. in Applied Physics (conferred June 2026)

September 2025 – Present

Pasadena, CA

Tsinghua University

B.S. in Mathematics and Physics, GPA 3.87/4.00

August 2021 – June 2025

Beijing, China

RESEARCH FOCUS

Integrated photonics; microresonator frequency combs; optical frequency division; low-noise microwave synthesis; precision timing.

RESEARCH PUBLICATIONS

Family of magnetic field-boosted superconductors in rhombohedral graphene.

J. Seo, A. A. Cotten, S. Ye, M. Xu, O. S. Sedeh, H. Weldeyesus, T. Han, Z. Lu, Z. Wu, **W. Xu**, J. Yang, E. Aitken, P. P. Liang, P. Pattanajakvijit, Z. Hadjri, R. Gazizulin, K. Watanabe, T. Taniguchi, M. Li, D. M. Zumbühl, and L. Ju.

Nature, 10.1038/s41586-026-10815-x (2026).

Signatures of chiral superconductivity in rhombohedral graphene.

T. Han, Z. Lu, Z. Hadjri, L. Shi, Z. Wu, **W. Xu**, Y. Yao, A. A. Cotten, O. S. Sedeh, H. Weldeyesus, J. Yang, J. Seo, S. Ye, M. Zhou, H. Liu, G. Shi, Z. Hua, K. Watanabe, T. Taniguchi, P. Xiong, D. M. Zumbühl, L. Fu, and L. Ju.

Nature, 643, 654–661 (2025).

Ballistic PbTe nanowire devices.

Y. Wang, F. Chen, W. Song, Z. Geng, Z. Yu, L. Yang, Y. Gao, R. Li, S. Yang, W. Miao, **W. Xu**, Z. Wang, Z. Xia, H.-D. Song, X. Feng, T. Wang, Y. Zang, L. Li, R. Shang, Q. Xue, K. He, and H. Zhang.

Nano Letters, 23, 11137–11144 (2023).

RESEARCH EXPERIENCE

California Institute of Technology

Graduate Researcher

September 2025 - Present

- **Integrated Photonics, Optical Frequency Combs & Microwave Synthesis (April 2026 - Present; Advisor: Prof. Kerry Vahala):** Design, simulate, lay out, nanofabricate, and characterize integrated photonic devices and microresonator chips for octave-spanning frequency-comb generation and ultra-low-noise microwave synthesis.
- Develop auxiliary-ring microresonator architectures for on-chip f-2f self-referencing and perform broadband optical and microwave characterization for compact precision-timing systems.

Superconductivity in Rhombohedral Graphene & Quantum Capacitance

Visiting Student Researcher, MIT (Advisor: Prof. Long Ju)

June 2024 - March 2025

- Fabricated high-quality, dual-graphite-gated rhombohedral graphene devices, contributing to research on chiral superconductivity published in *Nature*.
- Developed an ultralow-temperature quantum-capacitance measurement setup and an on-chip bilayer-graphene cryogenic amplifier for characterizing strongly correlated electronic states.

Supercurrent in Quantum Anomalous Hall Systems

Research Assistant, Tsinghua University (Advisor: Prof. Yayu Wang)

June 2023 - June 2024

- Induced superconductivity on the surface of Cr-doped Bi(Sb,Te) quantum anomalous Hall thin films and designed Josephson-junction geometries toward topological superconductivity.
- Reduced the junction gap length from 200 nm to 50 nm, enabling the observation of supercurrent in the system.
- Improved device quality with an Al₂O₃ layer protecting the films from electron-beam damage and a selective etch for the contacts.

AWARDS

Overseas Academic Research Support Program, Tsinghua (2024); ‘Qiyao’ Program Fellowship, Beijing Natural Science Foundation (2023); Comprehensive Excellence Scholarship, Tsinghua (2023); Spark Innovation Fellowship, top 1% at Tsinghua (2023); Chi-Sun Yeh Physics Academy Scholarship (2021–2024).